



Industrial Attachment – Day 5 Report

Data Preprocessing, System Setup & Initial Model Training | 9 September 2026 | MySoft Heaven BD

Team: AnomalyX

Team Members		
Sumaiya Zaman Student ID: 2104110	Karnel Chakma Student ID: 2104132	Joyeta Barua Moni Student ID: 2104130

CEO	Md. Mofakharul Islam	MySoft Heaven BD
HR	Aleya Parven	Human Resources

Date	9 September 2026	Day	Day 5
Company	MySoft Heaven BD	Session	Data Preprocessing & Initial Development
Supervised By			
Imdadul Haque		Sr. Software Engineer	
M. H. A. Afif		IT Business Analyst	

1. Dataset Cleaning and Preprocessing

On Day 5, we started preparing our synthetic dataset for actual use in the project. We cleaned the dataset and carried out the initial preprocessing steps to make the data more suitable for analysis and machine learning. We also worked on normalization and other preparation tasks required before training the models.

2. Data Preparation for Machine Learning

After cleaning the data, we prepared the relevant features for the initial machine learning experiments. The purpose of this stage was to make the dataset consistent and usable for training while also getting a better understanding of the transaction patterns and behavior represented in the data.

3. Frontend Development Setup

Alongside the data preparation work, we started setting up the frontend structure of our platform. We used **Next.js** as the frontend framework and **Tailwind CSS** for the user interface styling. The initial setup provided the foundation for developing the dashboard and other user-facing parts of the financial risk platform.

4. Backend Development Setup

We also began setting up the backend environment for connecting the application with the machine learning components. We used **FastAPI** for the backend/API layer. This initial setup was intended to provide a structure through which the frontend and future AI/ML services could communicate.

5. Initial AI/ML Model Training

After the initial data preparation, we trained some machine learning models to get an early view of the behavior of our dataset. The AI/ML models and tools used at this stage included **Scikit-learn, XGBoost, and Isolation Forest**. These were implemented initially to observe how the data behaved, explore the model outputs, and establish a starting point for the later fraud detection and anomaly detection pipeline.

6. Initial Observation and Project Direction

The initial model training was not treated as the final model evaluation. Instead, it helped us understand the dataset and obtain an early indication of how different approaches could work with our data. The results and observations from this stage would guide the next steps in feature engineering, model development, anomaly detection, and system integration.

7. Overall Reflection

Day 5 marked the transition from planning and dataset preparation toward actual implementation. By cleaning and preprocessing the data, setting up the frontend and backend environments, and running the first AI/ML models, we established the initial technical foundation of our project. This gave us a clearer direction for the development and modeling work that would follow.